

# Introduction to Astronomy and Space Science

PH409

IIT Gandhinagar

2025-26 Semester- II

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## Outline of the contents :

Based on the detailed list of topics in the general announcement, we plan to cover the following broad areas

1. Overview of the observable sky; planetary and satellite motion
2. Observational techniques and major worldwide facilities. Challenges of space exploration.
3. Stars and galaxies : General features of stellar formations; physical mechanisms and dynamics.
4. Cosmology - late Universe : The expanding Universe. General Relativity as the basis. Cosmic Microwave Background, Dark Matter, Dark energy. Gravitational waves.
5. Cosmology - the early Universe : Matter-anti-matter asymmetry; Inflationary Universe.

## Text Books/References :

1. An Introduction to Astronomy and Astrophysics, Pankaj Jain, CRC Press, 2024
2. The Physical Universe: An Introduction to Astronomy, Frank Shu, University Science Books; 1st edition, 1981
3. An Introduction to Modern Astrophysics, Carroll and Ostlie, Cambridge University Press; 2nd edition
4. Solar System Dynamics, Carl D. Murray, Cambridge University Press 2012.
5. Lectures on Astrophysics, S. Weinberg, Cambridge University Press, 2020

**Evaluation** : Exams I and II respectively 20 + 40 marks.

25 marks for assignments and 15 marks for class participation cum attendance.

15 marks total should be obtained in the 2 exams to ensure a passing grade.

**Attendance** : It is expected that you maintain regularity in attending. Attendance will be taken occasionally.

